

Problem 3.2

If \$100 is deposited into an account, whose accumulation function is

$$a(t) = 1 + 0.03\sqrt{t},$$

at time 0, find the effective rate for the first period (between time 0 and time 1) and second period (between time 1 and time 2).

Solution

For the first period, the effective rate is

$$i_{[0,1]} = \frac{a(1) - a(0)}{a(0)}.$$

Since

$$a(1) = 1 + 0.03\sqrt{1} = 1.03, \quad \text{and} \quad a(0) = 1 + 0.03\sqrt{0} = 1,$$

we get

$$i_{[0,1]} = \frac{1.03 - 1}{1} = 0.03 = 3\%.$$

For the second period, the effective rate is

$$i_{[1,2]} = \frac{a(2) - a(1)}{a(1)}.$$

Now,

$$a(2) = 1 + 0.03\sqrt{2}, \quad \text{and} \quad a(1) = 1.03,$$

so

$$i_{[1,2]} = \frac{(1 + 0.03\sqrt{2}) - 1.03}{1.03}.$$

Therefore,

$$i_{[1,2]} \approx 0.0126 = 1.26\%.$$

Hence, the effective rates are

$i_{[0,1]} = 3\%, \quad i_{[1,2]} \approx 1.26\%.$
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